InstaSHAP Reproducibility Report

Objective: reproduce the key InstaSHAP tabular experiments from the ICLR 2025 paper.

Datasets: Bike Sharing (synergy), Covertype (redundancy), Adult Income (supplementary).

Methodology: black-box baseline, additive GAMs, masked surrogate, and InstaSHAP Eq. (20).

Data source implementation: ucimlrepo fetch_ucirepo(id=275 / 31 / 2).

Outputs: metrics tables, plots, SHAP vs InstaSHAP comparison, and paper-reference tables.

Adult Overview

Dataset: adult

Task: classification

Features: age, fnlwgt, capital_gain, capital_loss, hours_per_week, workclass, education, marital_status, occupation, relation

Interaction pairs: []

Implementation notes:

- Black-box model: MLP by default, with optional RandomForest.

- GAM-1/GAM-2: additive neural subnetworks over feature groups.

- InstaSHAP: masked additive training against a learned masked surrogate.

- SHAP: permutation explainer aggregated back to original feature groups.

Adult Metrics

model	accuracy	log_loss	training_seconds	inference_seconds_mean
blackbox	0.8400	0.3685	0.3684	0.0011
gam1	0.8362	0.3750	1.9384	0.0089
instashap	0.8400	0.4235	2.3097	0.0066

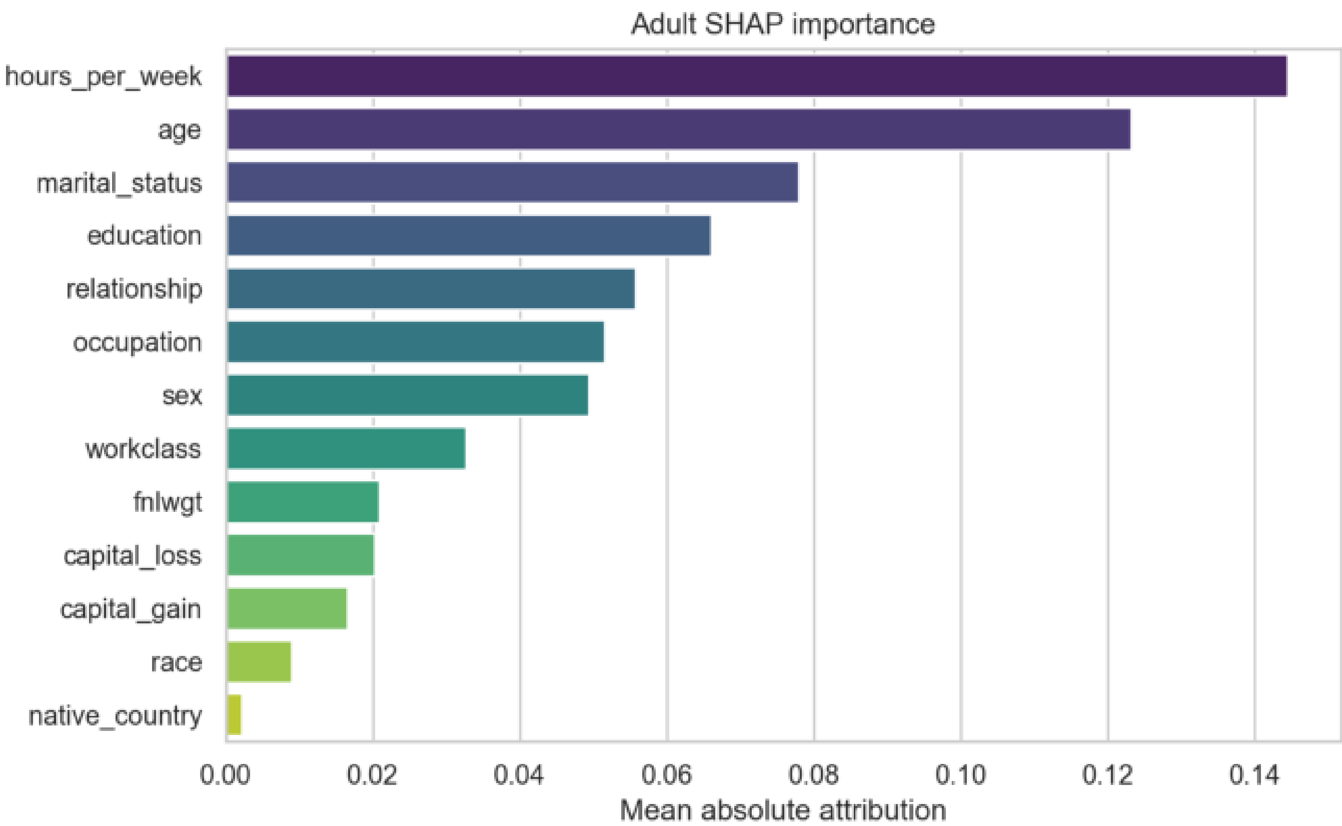
Adult vs Paper

model	reproduced	paper
gam1	0.8362	0.842
instashap	0.8400	0.843

Adult Explanation Comparison

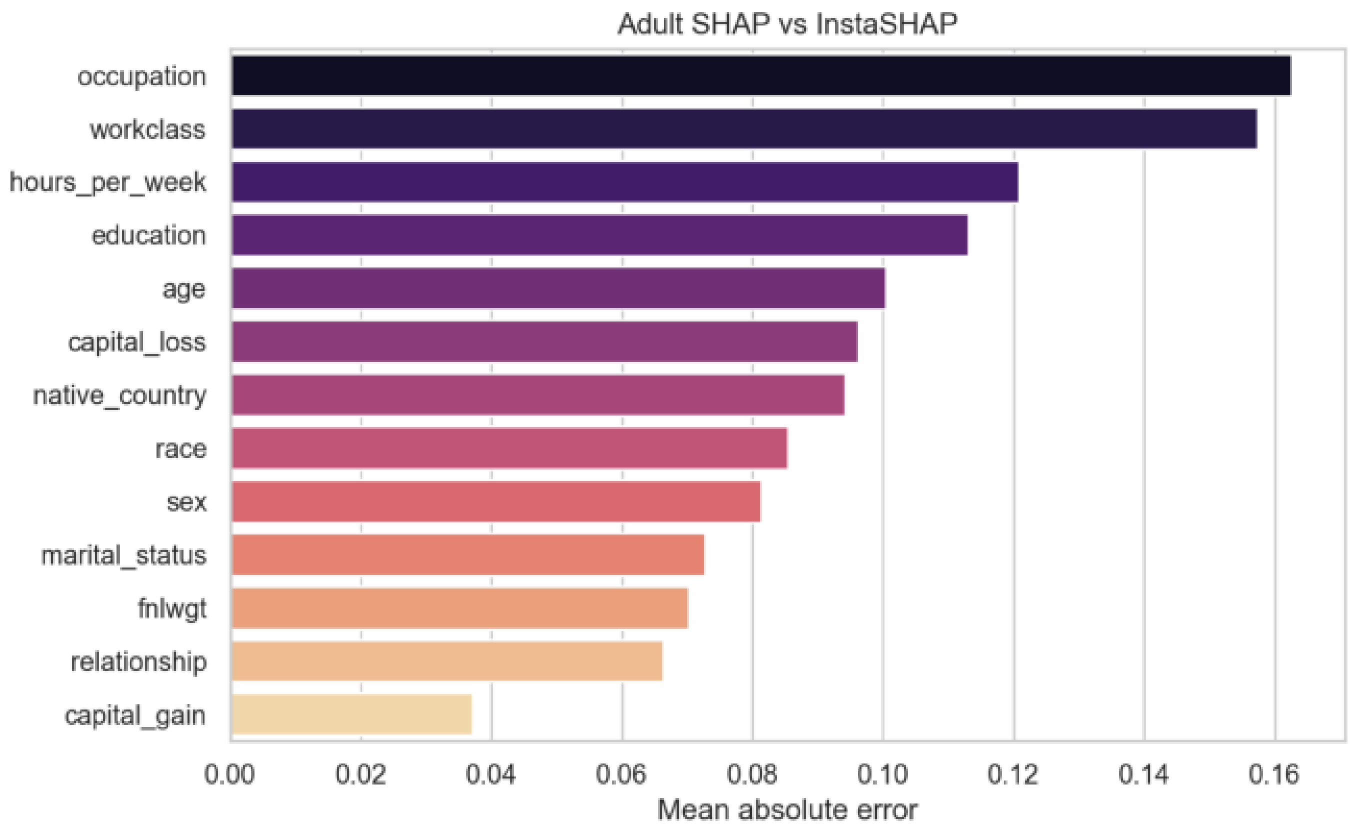
model	training_seconds	seconds_total	samples	mse	mae
surrogate	1.2105	NaN	NaN	NaN	NaN
shap	NaN	0.5953	24.0	NaN	NaN
instashap	NaN	0.0062	24.0	NaN	NaN
shap_vs_instashap	NaN	NaN	NaN	0.012	0.0968

Adult Plot



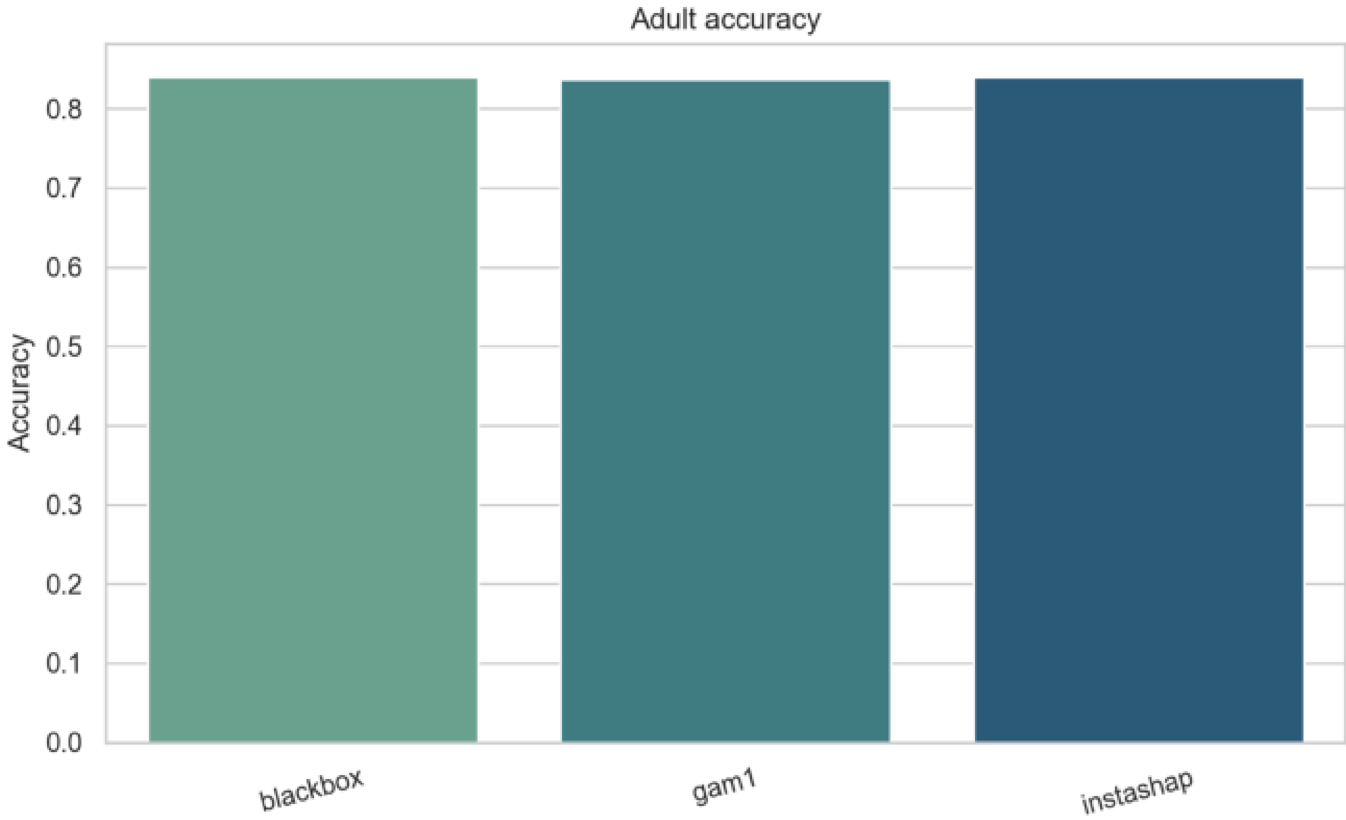
Adult SHAP importance
hours_per_week
age
marital_status
education
relationship
occupation
sex
workclass
fnlwgt
capital_loss
capital_gain
race
native_country
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Mean absolute attribution

Adult Plot

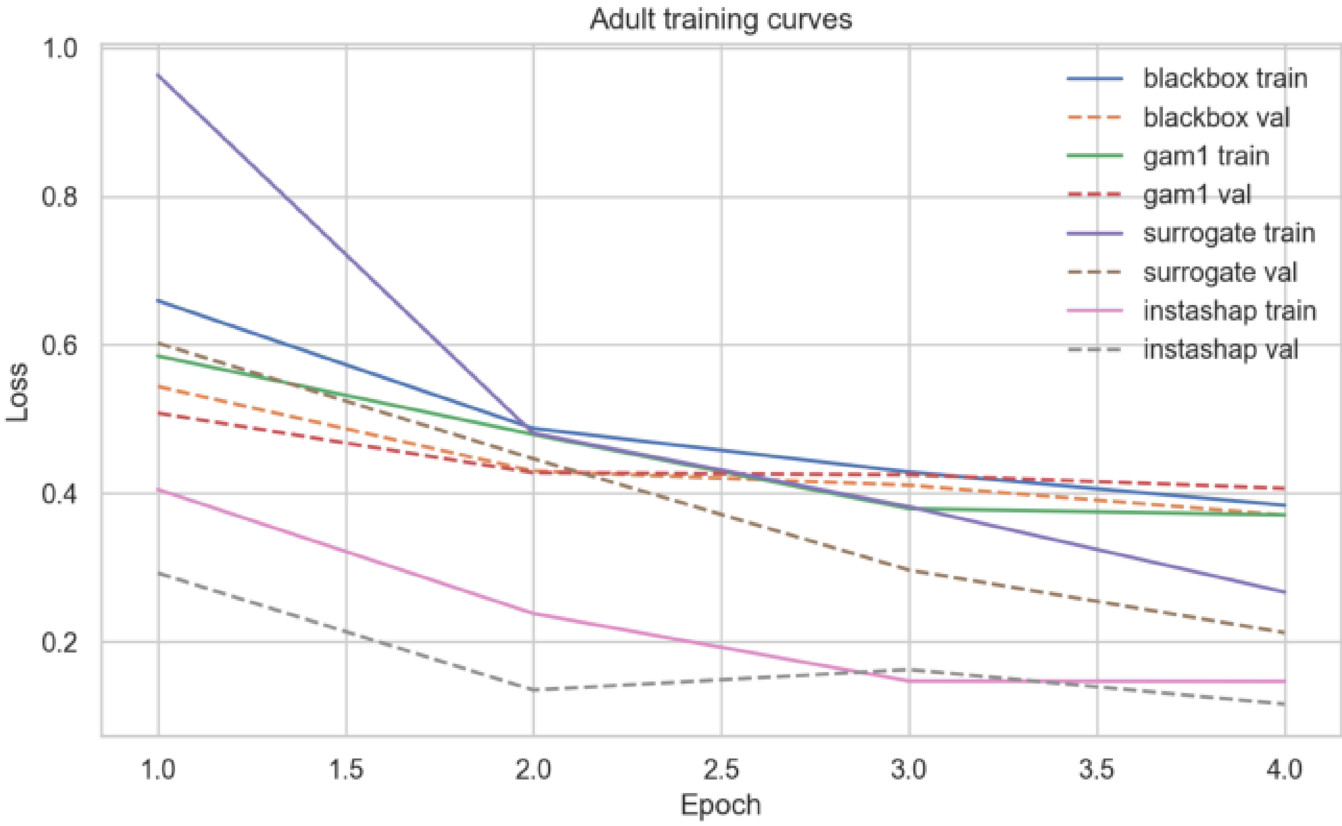


Adult SHAP vs InstaSHAP
occupation
workclass
hours_per_week
education
age
capital_loss
native_country
race
sex
marital_status
fnlwgt
relationship
capital_gain
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Mean absolute error

Adult Plot


Adult accuracy
Accuracy
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
blackbox
gam1
instashap

Adult Plot



Adult training curves
1.0
0.8
0.6
0.4
0.2
Loss
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Epoch
blackbox train
blackbox val
gam1 train
gam1 val
surrogate train
surrogate val
instashap train
instashap val

Bike Overview

Dataset: bike

Task: regression

Features: day_of_month, temp, atemp, hum, windspeed, season, year, month, hour, holiday, weekday, workingday, weather_

Interaction pairs: [['hour', 'workingday']]

Implementation notes:

- Black-box model: MLP by default, with optional RandomForest.

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- SHAP: permutation explainer aggregated back to original feature groups.

Bike Metrics

model	rmse	mse	r2	nmse_pct	training_seconds	inference_seconds_mean
blackbox	265.3422	70406.4766	-1.0485	204.8466	6.2930	0.0010
gam1	254.7480	64896.5312	-0.8882	188.8155	2.5011	0.0088
gam2	253.8135	64421.2734	-0.8743	187.4328	2.4362	0.0139
instashap	264.7821	70109.5781	-1.0398	203.9828	3.2291	0.0108

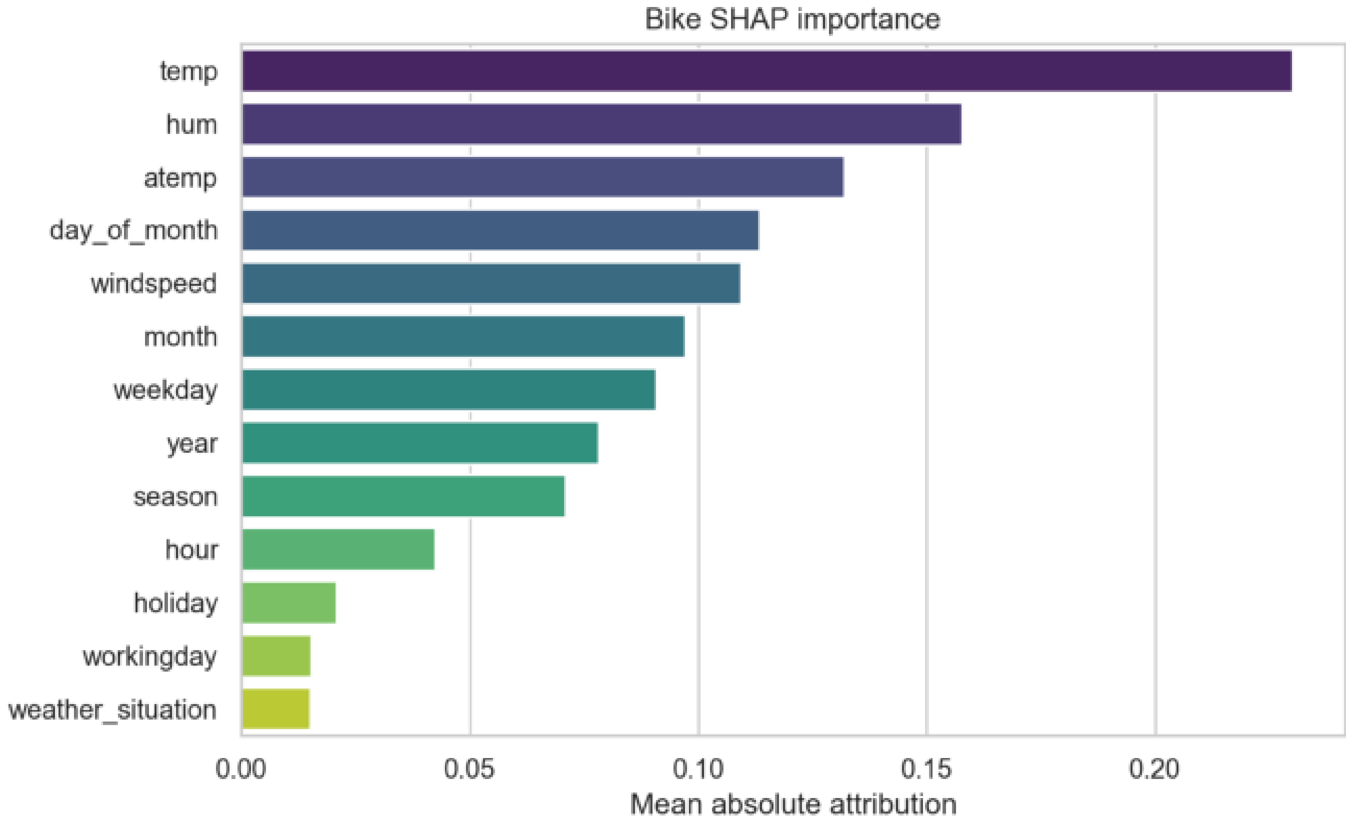
Bike vs Paper

model	reproduced	paper
blackbox	204.8466	6.59
gam1	188.8155	17.40
gam2	187.4328	6.23

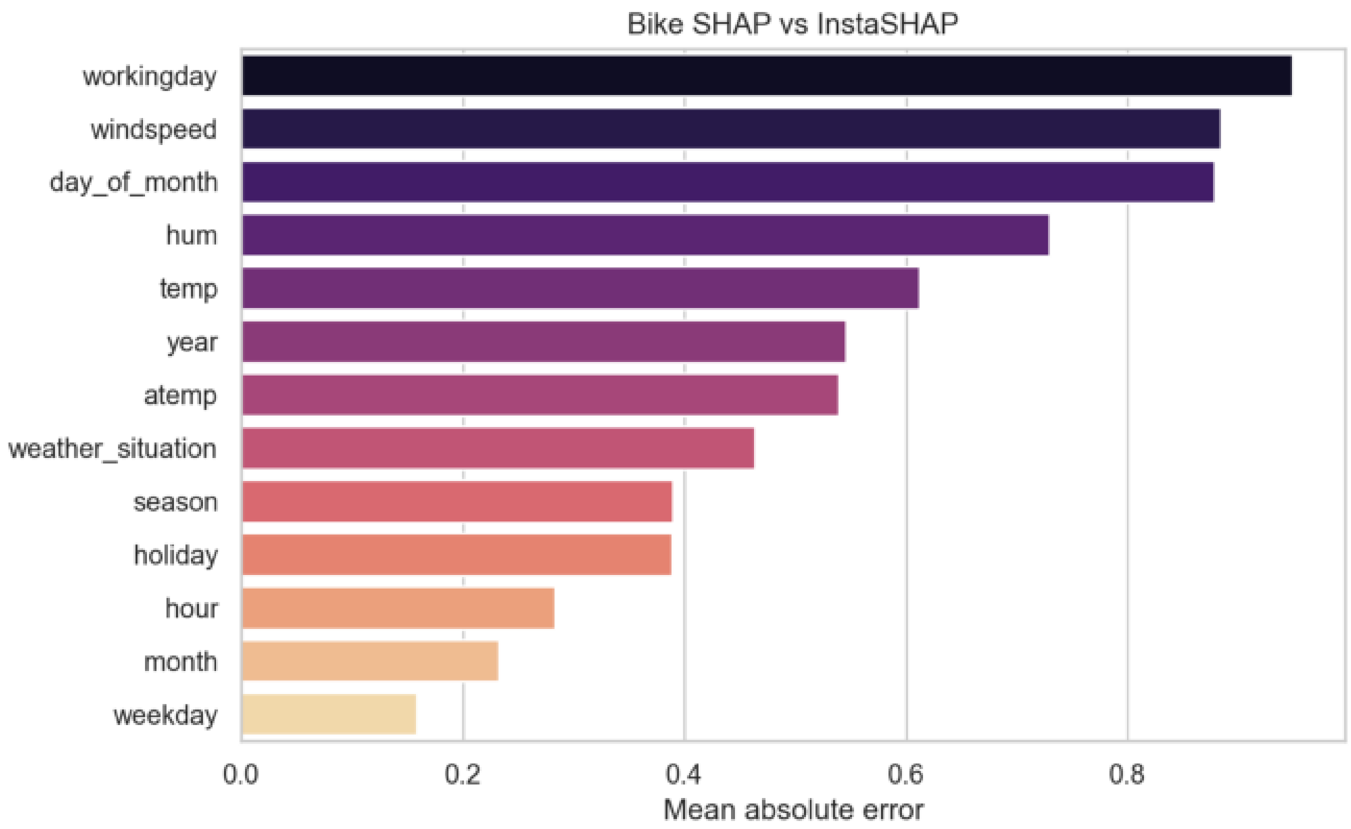
Bike Explanation Comparison

model	training_seconds	seconds_total	samples	mse	mae
surrogate	1.706	NaN	NaN	NaN	NaN
shap	NaN	15.1041	32.0	NaN	NaN
instashap	NaN	0.0082	32.0	NaN	NaN
shap_vs_instashap	NaN	NaN	NaN	0.3715	0.5427

Bike Plot

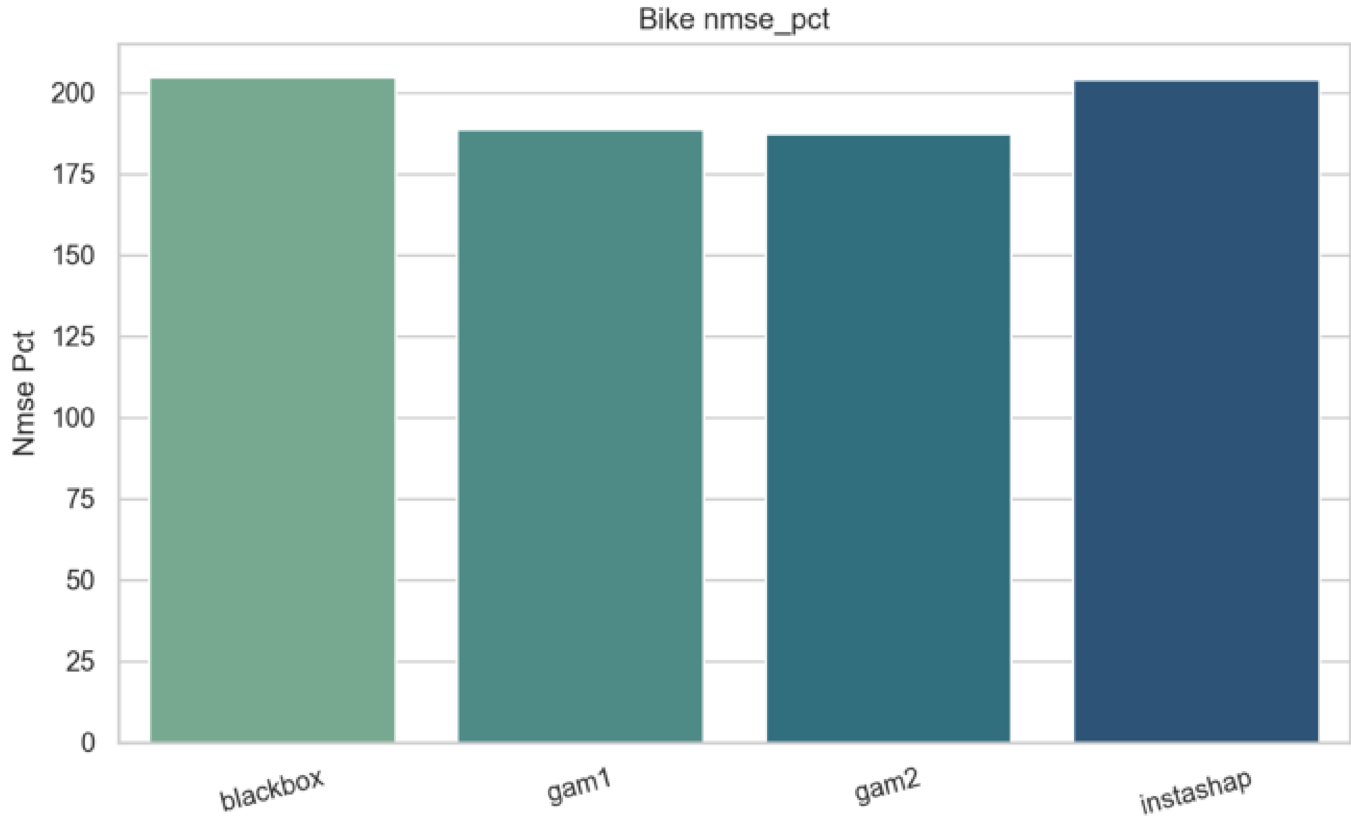

Bike SHAP importance
temp
hum
atemp
day_of_month
windspeed
month
weekday
year
season
hour
holiday
workingday
weather_situation
0.00
0.05
0.10
0.15
0.20
Mean absolute attribution

Bike Plot



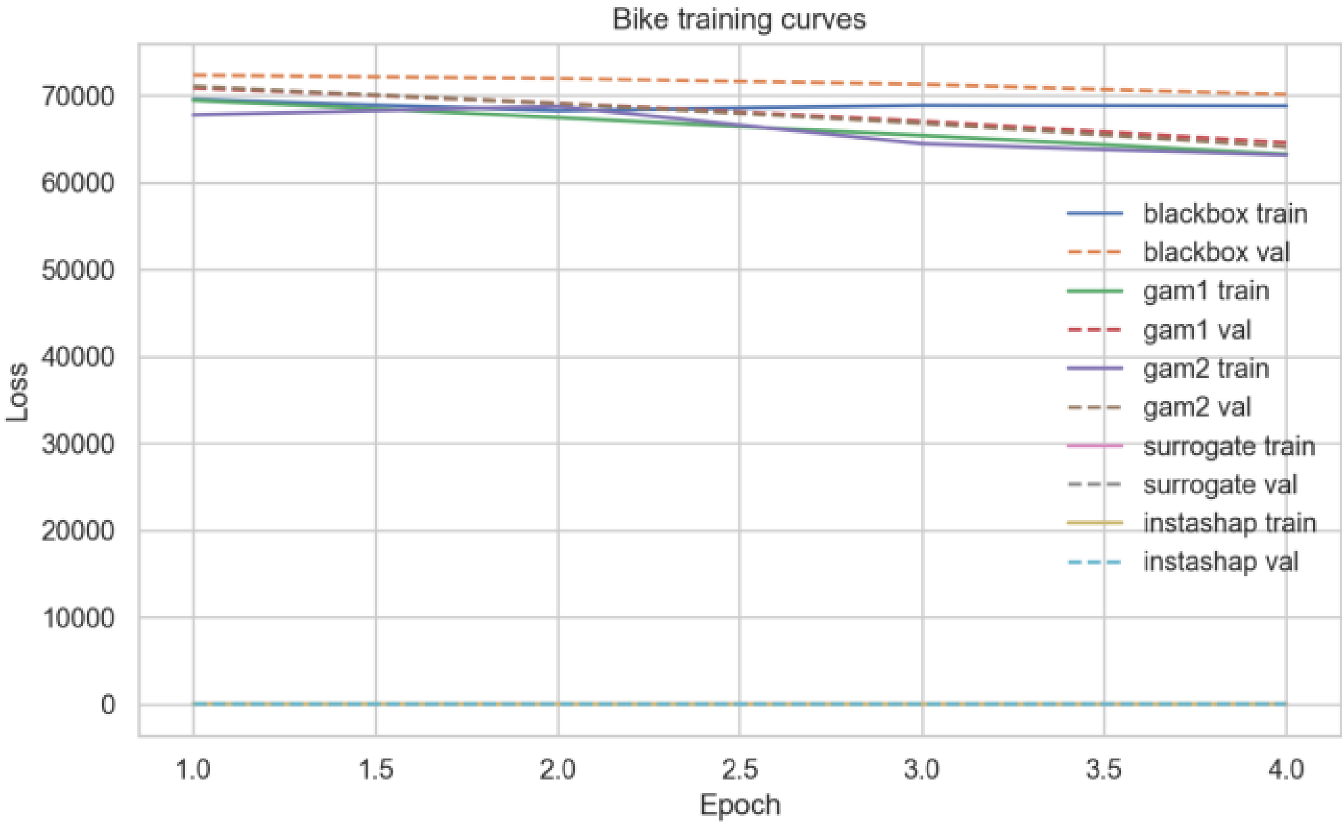
Bike SHAP vs InstaSHAP
workingday
windspeed
day_of_month
hum
temp
year
atemp
weather_situation
season
holiday
hour
month
weekday
0.0
0.2
0.4
0.6
0.8
Mean absolute error

Bike Plot



Bike nmse_pct
Nmse Pct
0
25
50
75
100
125
150
175
200
blackbox
gam1
gam2
instashap

Bike Plot



Bike training curves
Loss
Epoch
70000
60000
50000
40000
30000
20000
10000
0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
blackbox train
blackbox val
gam1 train
gam1 val
gam2 train
gam2 val
surrogate train
surrogate val
instashap train
instashap val

Covertype Overview

Dataset: covertype

Task: classification

Features: elevation, aspect, slope, horizontal_distance_to_hydrology, vertical_distance_to_hydrology, horizontal_distance_to

Interaction pairs: [['elevation', 'soil_climate_zone']]

Implementation notes:

- Black-box model: MLP by default, with optional RandomForest.
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Covertype Metrics

model	accuracy	log_loss	training_seconds	inference_seconds_mean
blackbox	0.6375	0.9931	0.7191	0.0010
gam1	0.6688	0.8895	1.7398	0.0108
gam2	0.6912	0.8041	1.3748	0.0088
instashap	0.4875	1.3345	1.9814	0.0057

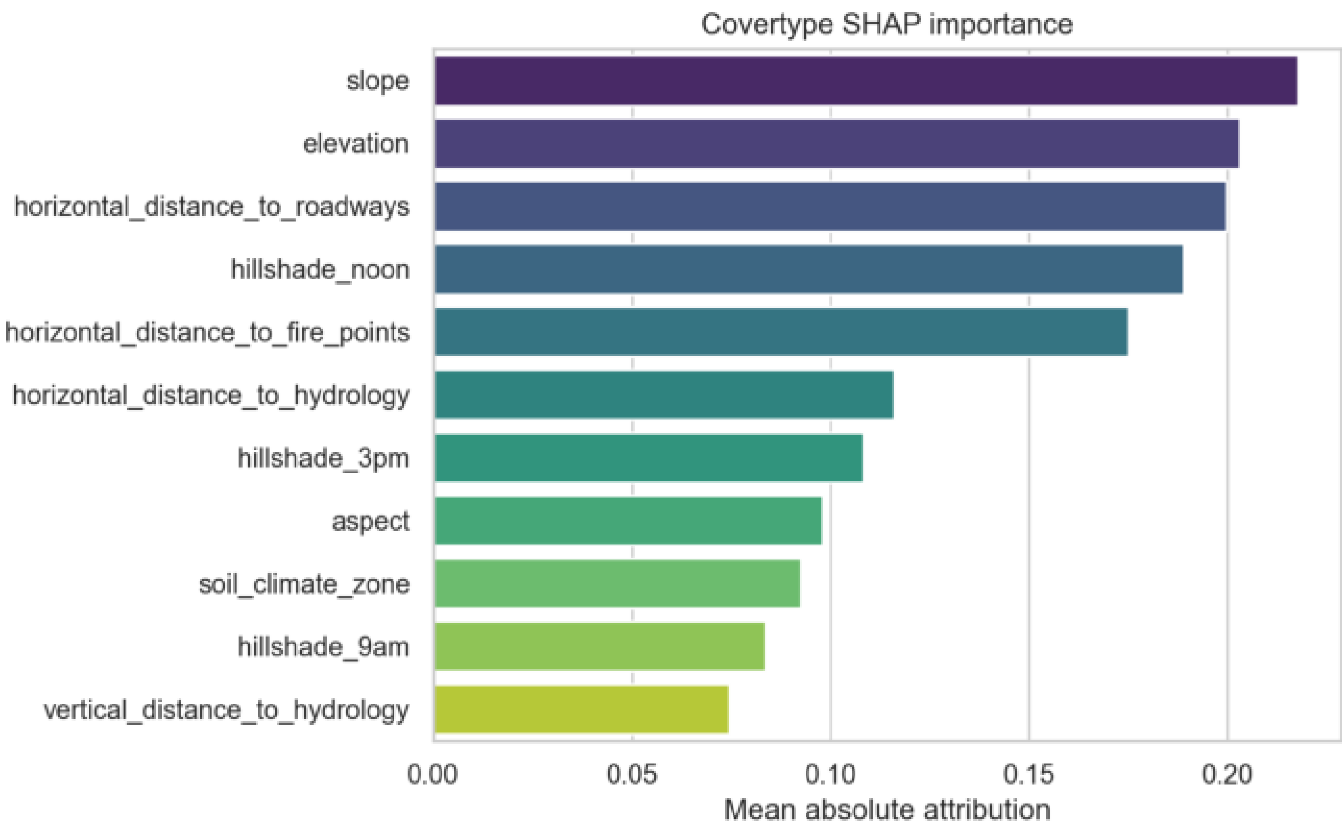
Covertype vs Paper

model	reproduced	paper
blackbox	0.6375	0.804
gam1	0.6688	0.724
gam2	0.6912	0.822

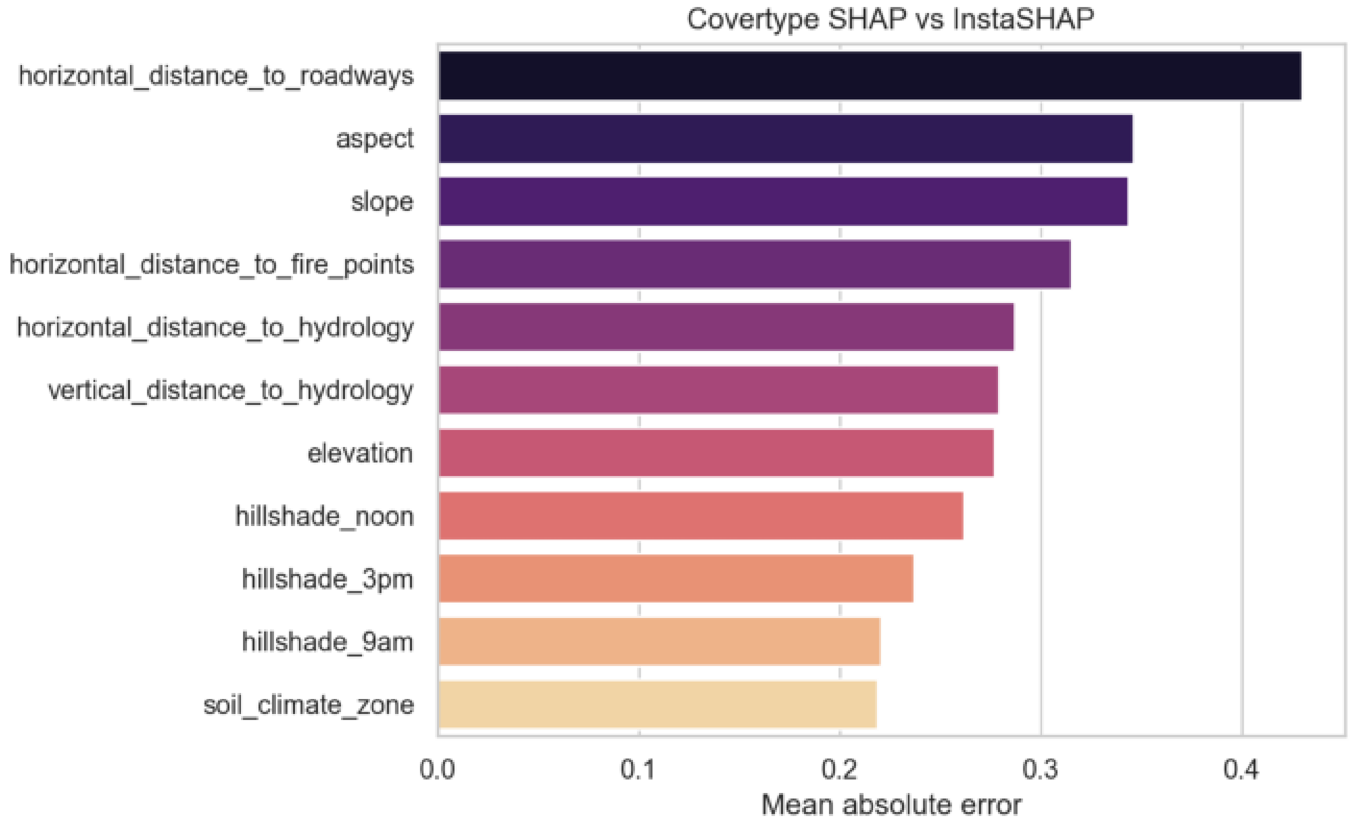
Covertype Explanation Comparison

model	training_seconds	seconds_total	samples	mse	mae
surrogate	1.0807	NaN	NaN	NaN	NaN
shap	NaN	0.8008	24.0	NaN	NaN
instashap	NaN	0.0058	24.0	NaN	NaN
shap_vs_instashap	NaN	NaN	NaN	0.1101	0.2922

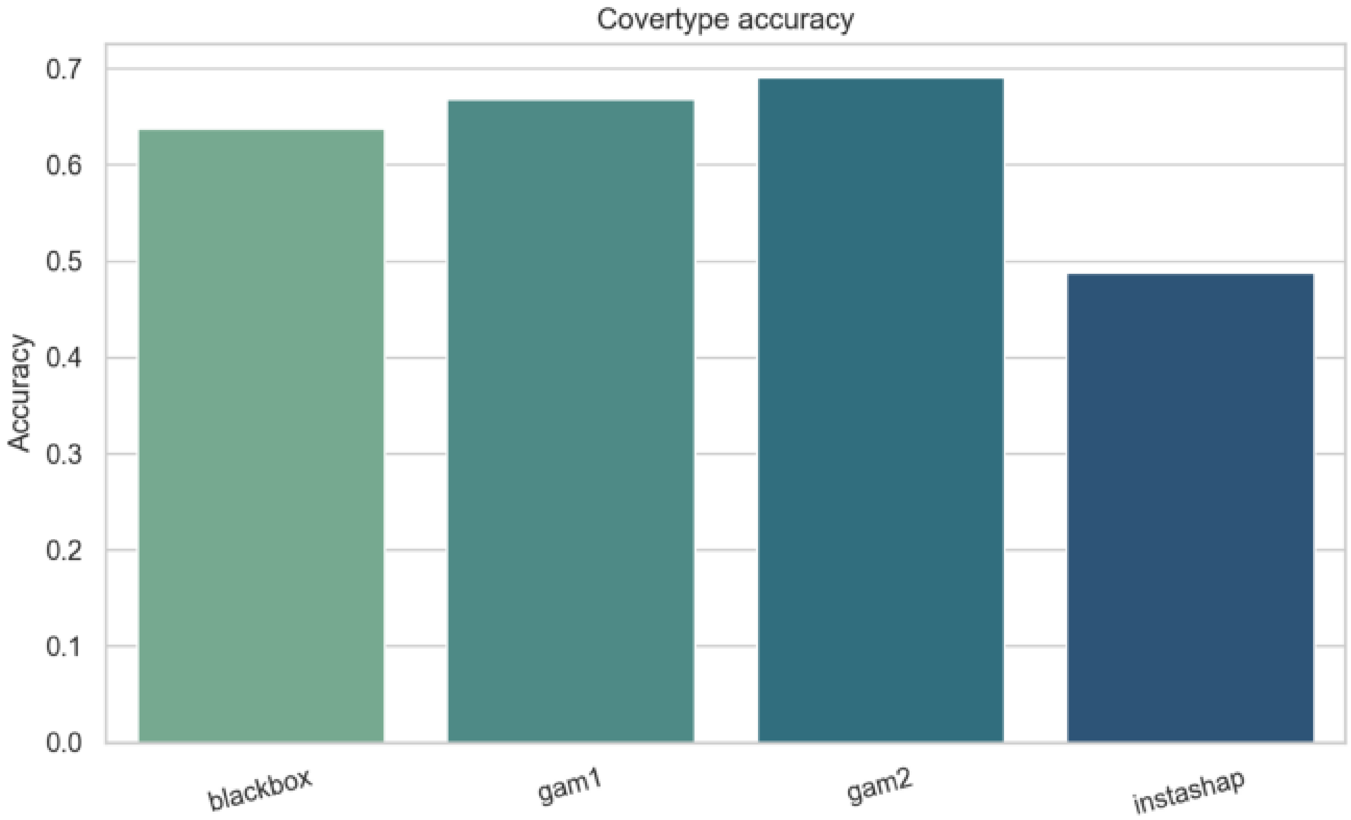
Covertype Plot



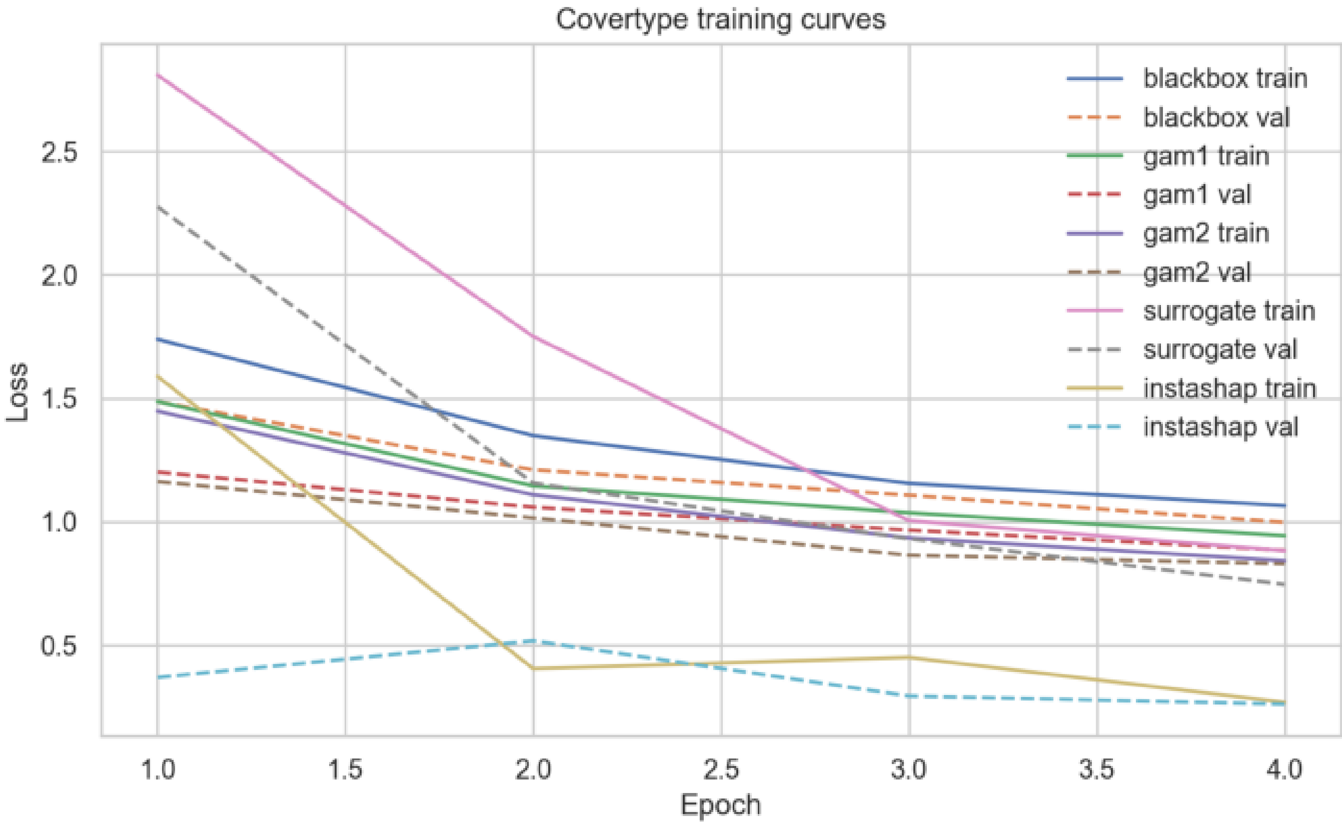
Covertype Plot



Covertype Plot



Covertype Plot



Comparison and Observations

Observations:

- The report juxtaposes reproduced metrics against the values quoted in the paper.
- Remaining gaps usually come from limited epochs, smaller subsets, and missing original hyperparameters.
- InstaSHAP artifacts include both predictive performance and explanation-fidelity measurements.

Recommendation:

Increase epochs and Covertype sample size for a closer numerical match to the paper before final benchmarking.